

Panel MS-AR(1) with random-effects intercepts, full sample

Real GDP per worker (period-average USD)

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes:

$$y_{it} = a_i + g t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}), \quad (3)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2) \quad (\text{random effects}). \quad (4)$$

The outcome is log real GDP per worker (period-average USD) from the 2026-09-14 quarterly panel. Calendar time is taken from the period stamp t (year-fraction). The trend is linear in calendar time t (g per year, common across countries). In the pooled specification $a_i \equiv a$. In the random-effects specification the a_i are i.i.d. Normal draws; α and ω are estimated by maximum likelihood, integrating each country's Hamilton-filter likelihood by 11-point Gauss–Hermite quadrature. Cycles are plotted using the posterior mean of a_i . Latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ and ρ switch with the regime. The median μ is pinned at 0. $|\rho| < 0.99$. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters. π is the ergodic distribution of Π . $E[z]$ is the long-run mean of the cycle implied by μ , ρ , and Π .

2 Common intercept and linear trend

2026-09-14 panel, full sample: log real GDP per worker. 90 countries, 8489 observations, 1963-Q1–2026-Q2. Fitted countries: 90. Observations: 8489. Log-likelihood: 11645.37. Ergodic mean of the cycle $E[z] = 0.6666$ (0.1031). Common intercept $a = -5.5897$ (0.1284). Common slope $g = 0.0004$ (0.0018).

Table 1: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.5619 (0.4211)	0.0000 —	1.5340 (0.0919)
σ	0.2062 (0.0065)	0.0827 (0.0016)	0.0379 (0.0006)
ρ	0.9900	0.9900 (0.0000)	0.9900 (0.0000)
Π_0	0.9896 (0.0044)	0.0080 (0.0039)	0.0024 (0.0026)
Π_1	0.0021 (0.0009)	0.9883 (0.0023)	0.0096 (0.0021)
Π_2	0.0003 (0.0007)	0.0089 (0.0018)	0.9908 (0.0016)
π	0.1014 (0.0277)	0.4270 (0.0405)	0.4716 (0.0423)

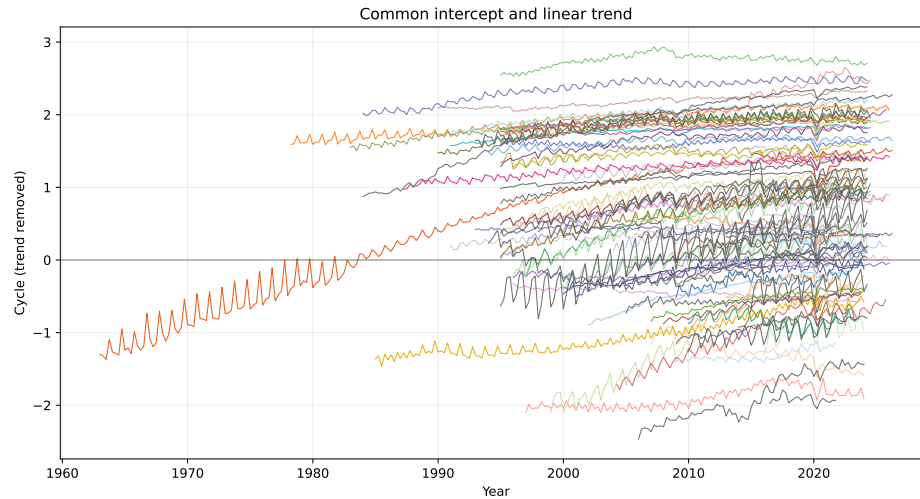


Figure 1: Country cycles after removing the linear trend.

3 Random-effects intercepts, common linear trend

2026-09-14 panel, full sample: log real GDP per worker. 90 countries, 8489 observations, 1963-Q1–2026-Q2. Fitted countries: 90. Observations: 8489. Log-likelihood: 11895.91. Ergodic mean of the cycle $E[z] = 0.0789$ (0.0372). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -6.5548$ (0.0471), $\omega = 0.6843$ (0.0114). Posterior-mean a_i : mean -6.253, min -8.516, max -4.582. Common slope $g = 0.0210$ (0.0008).

Table 2: Regime AR(1), transition matrix, and ergodic π . Columns are regimes.

	(0)	(1)	(2)
μ	-0.5191 (0.0280)	0.0000 —	0.3828 (0.0575)
σ	0.1817 (0.0054)	0.0821 (0.0014)	0.0371 (0.0006)
ρ	0.5041 (0.0391)	0.9271 (0.0079)	0.9900 (0.0002)
Π_0	0.9914 (0.0032)	0.0036 (0.0025)	0.0049 (0.0024)
Π_1	0.0020 (0.0008)	0.9911 (0.0019)	0.0068 (0.0018)
Π_2	0.0001 (0.0001)	0.0054 (0.0011)	0.9945 (0.0012)
π	0.0925 (0.0264)	0.3684 (0.0419)	0.5391 (0.0442)

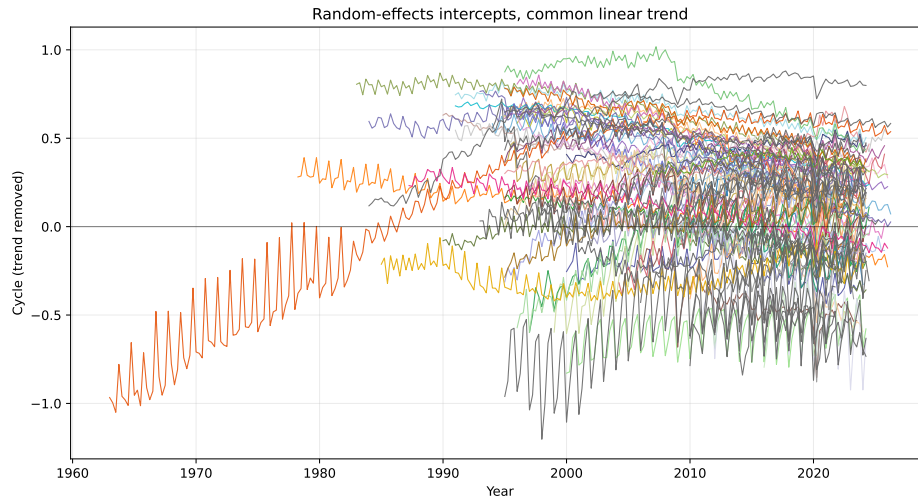


Figure 2: Country cycles after removing the linear trend.